

GridPulse Analytics

AI-Driven Load Forecasting for Renewable-Heavy Power Grids

The Problem: Renewable Grids Are Hard to Balance

As utilities add more solar and wind capacity, grid operators face increasingly volatile supply that legacy forecasting tools were never designed to handle. Poor load forecasting leads to costly curtailment of clean energy, emergency fossil-fuel peaker plant activation, and, in worst cases, regional instability.

Our Solution: Real-Time Grid Intelligence

GridPulse provides utilities with an AI-driven forecasting and load-balancing platform that ingests weather data, historical consumption, and live sensor feeds to predict supply-demand imbalances up to 48 hours in advance. Grid operators use our dashboard and API to pre-emptively dispatch storage and adjust renewable output, reducing curtailment and peaker-plant reliance (B2B SaaS model sold directly to utility operators).

Technology Advantage

- Ensemble forecasting models combining gradient-boosted trees and temporal graph neural networks, tuned per-region for grid topology.
- Custom time-series data structures for ingesting and querying millions of sensor readings per minute with low-latency aggregation.
- Continuous backtesting pipeline that benchmarks forecast accuracy against realized grid data to retrain models automatically as seasonal patterns shift.

Python & C++ · Temporal Graph Neural Networks · Real-Time Sensor Ingestion

\$6B+ TARGET GRID-SOFTWARE MARKET	22% REDUCTION IN CURTAILMENT	B2B SaaS BUSINESS MODEL
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Proposal for Terra Nova Capital

We are seeking a **\$4,000,000 Series A** investment to expand our forecasting engine to two additional regional grid operators and grow our applied ML team.

This opportunity aligns with Terra Nova Capital's thesis of investing in software infrastructure that accelerates renewable energy adoption by solving the operational challenges utilities face at scale.